1 Hypergraph Shortest Hyperpath Hardness

Title. Directed Hypergraphs and Applications

Paper. Gallo, Longo, Pallottino, Nguyen (1993). `https://static.aminer.org/pdf/PDF/000/836/884/directed_hypergraphs_and_applications.pdf`

Model and algorithmic setting. The paper studies shortest path problems in directed hypergraphs, where a hyperedge may connect multiple tail vertices to multiple head vertices. The authors define the notion of a B–hyperpath and study the problem of finding a minimum–weight B–hyperpath between two vertices.

Lower bound. They prove that computing a minimum–weight B–hyperpath is NP–hard: Shortest B–Hyperpath is NP–hard.

Relevance to our problem. If each player's edge set is encoded as a hyperedge, then feasible paths in the union graph correspond to hyperpaths. This result implies that any fully general formulation of our problem cannot admit a polynomial–time centralized algorithm unless P = NP, establishing a strong computational lower bound.

2 Distributed Shortest Path Lower Bounds

Title. Reachability and Shortest Paths in the Broadcast CONGEST Model

Paper. Chechik and Mukhtar (DISC 2019). `https://arxiv.org/abs/1910.05645`

Model and algorithmic setting. This work studies single–source shortest path (SSSP) in the Broadcast CONGEST distributed model, where each node initially knows only its incident edges and communication per round is limited to small messages.

Lower bound. They show that any exact shortest path algorithm requires at least $\widetilde{\Omega}(\sqrt{n}+D)$ communication rounds, where n is the number of vertices and D is the graph diameter.

Relevance to our problem. Our multi–player model can be viewed as a distributed shortest path computation in which each player initially holds only a subset of edges. This bound implies that even when shortest paths are easy to compute centrally, any interactive or distributed protocol must incur polynomial communication cost, ruling out highly efficient coordination among players.

3 Streaming and Multi–Pass Lower Bounds

Title. Near–Quadratic Lower Bounds for Two–Pass Graph Streaming Algorithms

Paper. Assadi and Raz (2023). https://arxiv.org/abs/2009.01161

Model and algorithmic setting. The paper studies s–t reachability and shortest path problems in the graph streaming model, where edges arrive as a stream and the algorithm is allowed only a small number of passes with limited memory.

Lower bound. They prove that any two–pass streaming algorithm for s–t reachability requires $n^{2-o(1)}$ space. Related results show that any semi–streaming algorithm achieving even a constant–factor approximation for shortest paths requires $\Omega\left(\frac{\log n}{\log\log n}\right)$ passes over the input.

Relevance to our problem. Viewing each player's edge set as a separate data stream, these bounds imply that discovering short paths in the union graph necessarily requires either near–quadratic memory or many rounds of interaction. This provides an information–theoretic barrier for protocols with limited communication or memory.